



B.Tech. CSE VI CSS 336
ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING LAB

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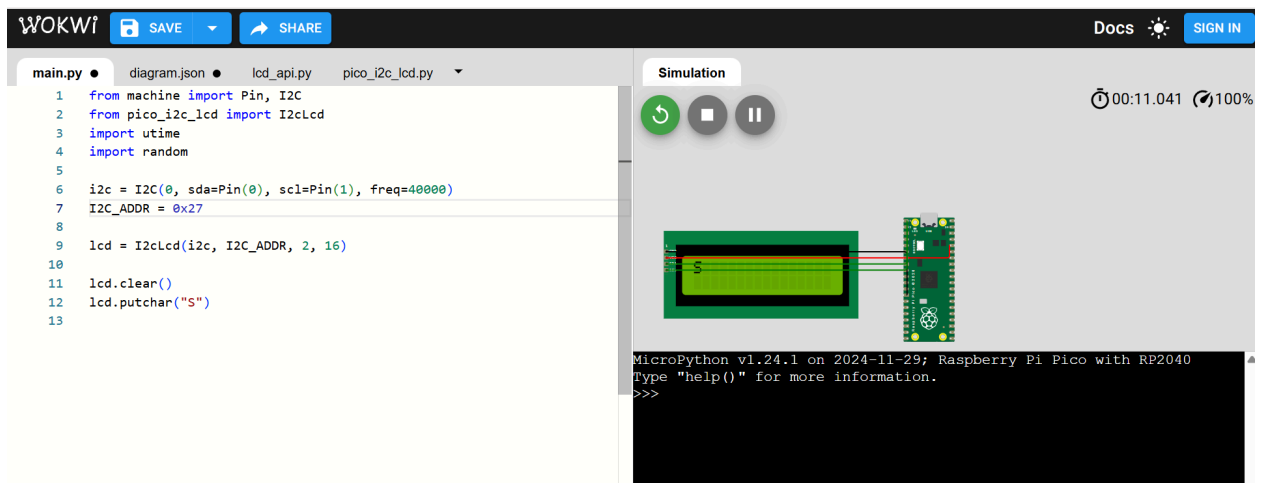
Roll Number: BTECH/10007/23

Embedded Systems Laboratory

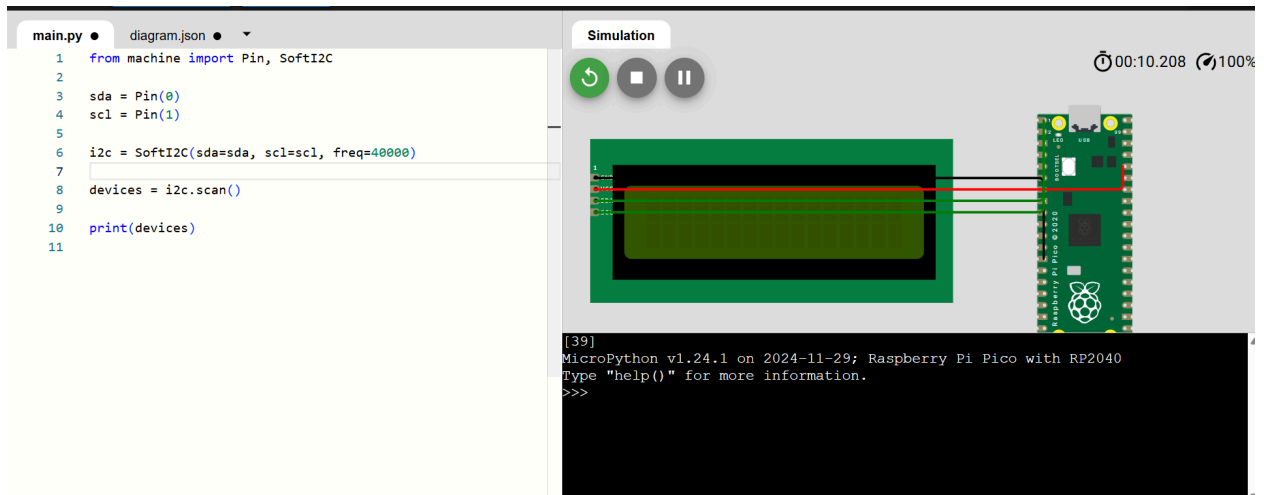
Assignment- III

February 4th, 2026

1. Connection establishment



Printing a String



1. Put a string on a I2C LCD panel

```
from machine import Pin, I2C
from pico_i2c_lcd import I2cLcd
import utime
import random
```

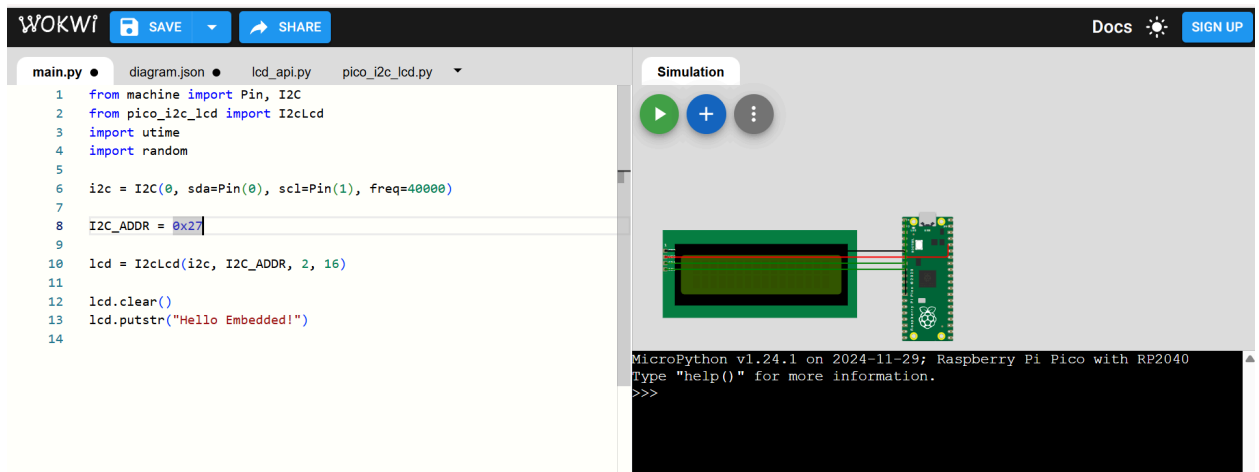
```
i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
```

```
I2C_ADDR = 0x27
```

```
lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
```

```
lcd.clear()
```

```
lcd.putstr("Hello Embedded!")
```



2. Display a running counter on the I2C panel, where the numbers up dates every 1 second

```
from machine import Pin, I2C
from pico_i2c_lcd import I2cLcd
import utime
import random
```

```
i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
```

```
I2C_ADDR = 0x27
```

```
lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
```

```
count = 0
```

```
lcd.clear()
```

```

while True:
    lcd.move_to(0, 0)
    lcd.putstr("Counter:")
    lcd.move_to(0, 1)
    lcd.putstr(str(count) + "    ")
    count += 1
    utime.sleep(1)

```

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main.py diagram.json lod_api.py pico_i2c_lcd.py

```

1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6
7 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
8
9
10 I2C_ADDR = 0x27
11
12 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
13
14 count = 0
15 lcd.clear()
16
17
18 while True:
19     lcd.move_to(0, 0)
20     lcd.putstr("Counter:")
21     lcd.move_to(0, 1)
22     lcd.putstr(str(count) + "    ")
23     count += 1
24     utime.sleep(1)

```

Simulation 00:02.865 24%

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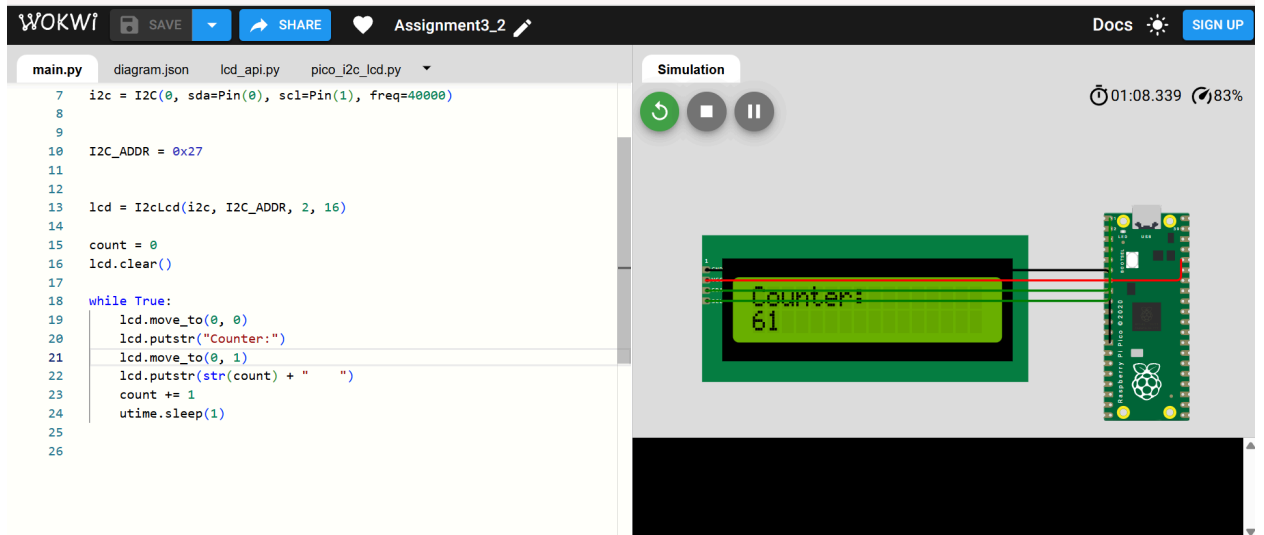
main.py diagram.json lod_api.py pico_i2c_lcd.py

```

1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6
7 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
8
9
10 I2C_ADDR = 0x27
11
12 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
13
14 count = 0
15 lcd.clear()
16
17
18 while True:
19     lcd.move_to(0, 0)
20     lcd.putstr("Counter:")
21     lcd.move_to(0, 1)
22     lcd.putstr(str(count) + "    ")
23     count += 1
24     utime.sleep(1)

```

Simulation 00:07.928 57%



3. Generate a random number between 1 to 5. Display the same on a LCD screen and light up a corresponding LED.

```
from machine import Pin, I2C
from pico_i2c_lcd import I2cLcd
import utime
import random
```

```
i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
I2C_ADDR = 0x27
```

```
lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
```

```
leds = [
    Pin(10, Pin.OUT),
    Pin(11, Pin.OUT),
    Pin(12, Pin.OUT),
    Pin(13, Pin.OUT),
    Pin(14, Pin.OUT)
]
```

```
for led in leds:
    led.off()
```

```

lcd.clear()
lcd.putstr("Random No:")

while True:
    for led in leds:
        led.off()

    num = random.randint(1, 5)

    lcd.move_to(0, 1)
    lcd.putstr(" ")
    lcd.move_to(0, 1)
    lcd.putstr(str(num))

    leds[num - 1].on()
    utime.sleep(2)

```

main.py • diagram.json • lcd_api.py • pico_i2c_lcd.py ▾

```

1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6
7
8 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
9 I2C_ADDR = 0x27
10
11
12 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
13
14

```

Simulation

00:13

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main.py • diagram.json • lcd_api.py • pico_i2c_lcd.py ▾

```

1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6
7
8 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
9 I2C_ADDR = 0x27
10
11
12 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
13
14
15 leds = [
16     Pin(10, Pin.OUT),
17     Pin(11, Pin.OUT),
18     Pin(12, Pin.OUT),
19     Pin(13, Pin.OUT),
20     Pin(14, Pin.OUT)
21 ]
22
23 for led in leds:
24     led.off()

```

Simulation

00:23.998 99%

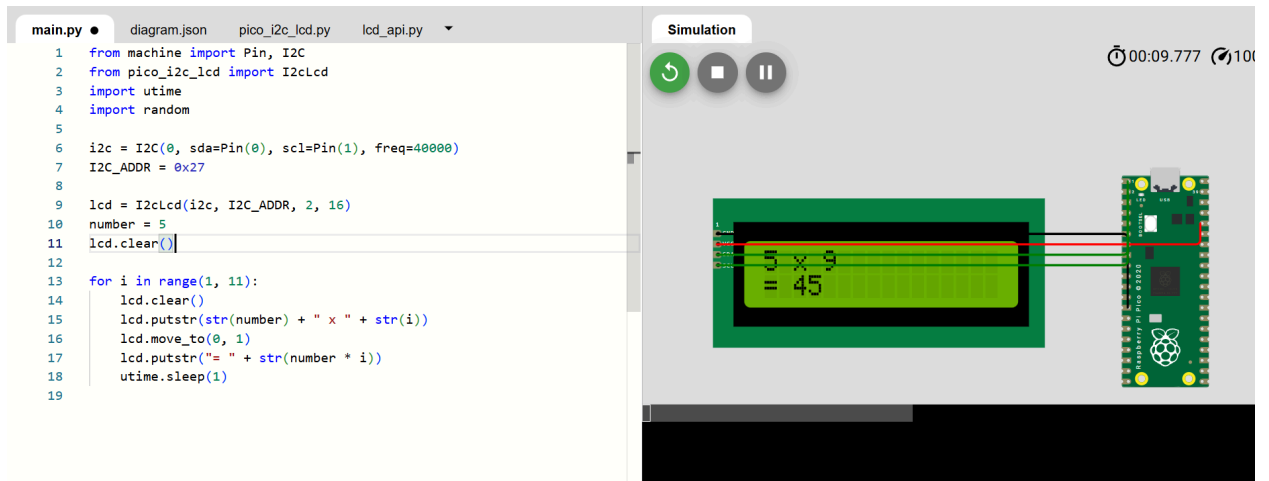
4. Print the multiplication table of a number line by line on a LCD panel.

```
from machine import Pin, I2C
from pico_i2c_lcd import I2cLcd
import utime
import random

i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
I2C_ADDR = 0x27

lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
number = 5
lcd.clear()

for i in range(1, 11):
    lcd.clear()
    lcd.putstr(str(number) + " x " + str(i))
    lcd.move_to(0, 1)
    lcd.putstr("=" + str(number * i))
    utime.sleep(1)
```



2. Print a line of scrolling text on LCD panel.

```

from machine import Pin, I2C
from pico_i2c_lcd import I2cLcd
import utime
import random

i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
I2C_ADDR = 0x27

lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
number = 5
lcd.clear()

text = "Embedded Systems Lab "
lcd.clear()

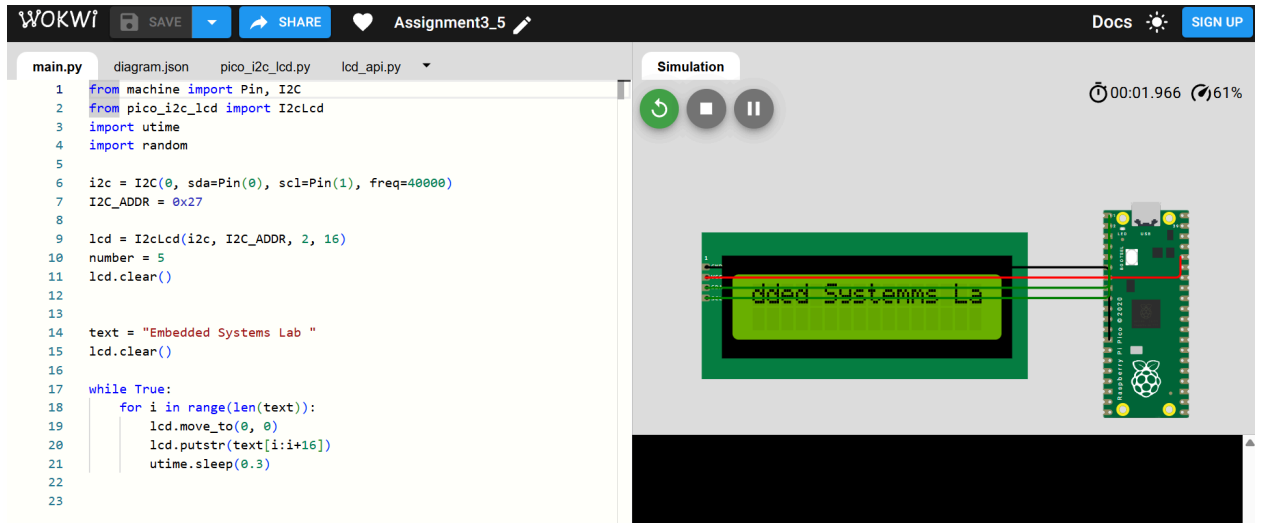
```

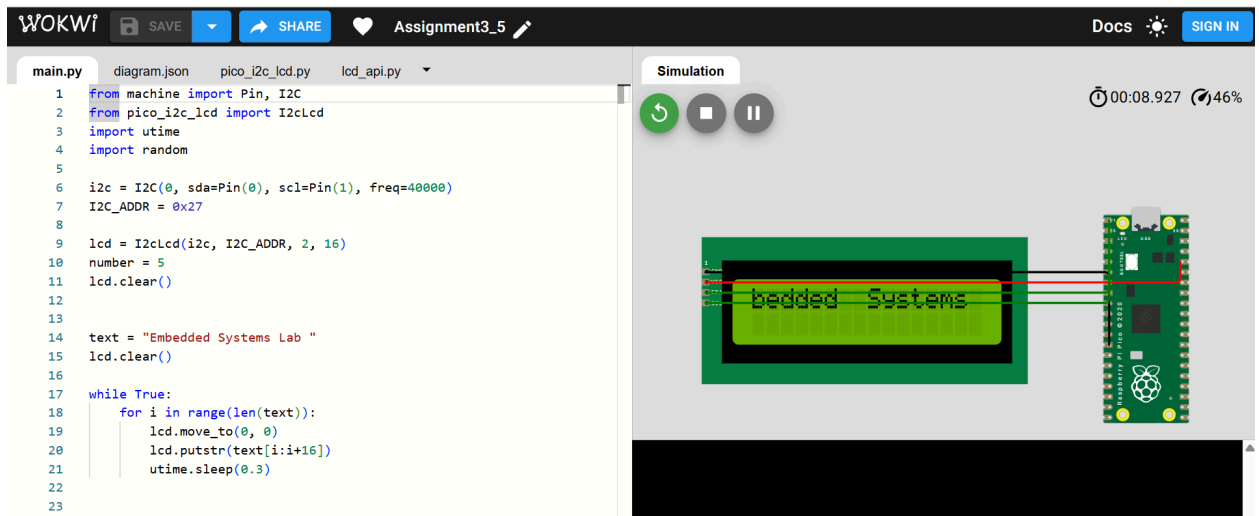
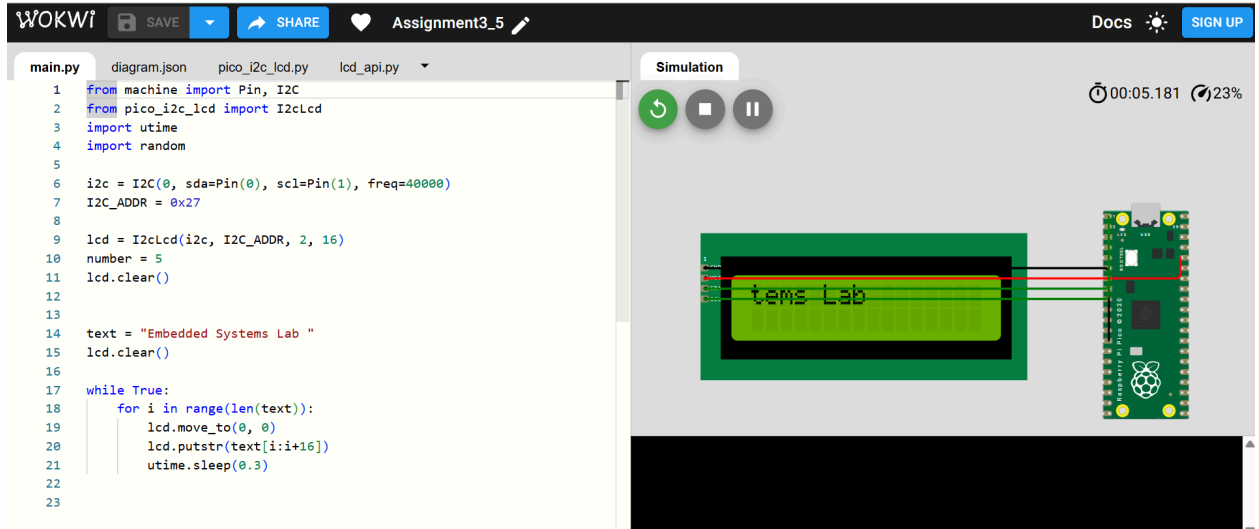


```

while True:
    for i in range(len(text)):
        lcd.move_to(0, 0)
        lcd.putstr(text[i:i+16])
        utime.sleep(0.3)

```





3. Simulate the bouncing ball game. The output of your program should how many jumps were required for the ball to land on the target lo cation. The game starts of with a target having a given co-rdinate i.e. (x,y). The ball is placed randomly at a point on the screen and then jumps about randomly. You win when the ball lands on the target square within N moves, else you lose the game

```

from machine import Pin, I2C
from pico_i2c_lcd import I2cLcd
import utime
import random

```

```

i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
I2C_ADDR = 0x27

```

```

lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)

```

```
number = 5
lcd.clear()

MAX_MOVES = 20
target_x = 10
target_y = 1

ball_x = random.randint(0, 15)
ball_y = random.randint(0, 1)

moves = 0
lcd.clear()

while moves < MAX_MOVES:
    lcd.clear()

    # draw target
    lcd.move_to(target_x, target_y)
    lcd.putstr("X")

    # draw ball
    lcd.move_to(ball_x, ball_y)
    lcd.putstr("O")

    if ball_x == target_x and ball_y == target_y:
        lcd.clear()
        lcd.putstr("YOU WIN!")
        lcd.move_to(0, 1)
        lcd.putstr("Moves: " + str(moves))
        break

    # random jump
    ball_x = random.randint(0, 15)
    ball_y = random.randint(0, 1)

    moves += 1
    utime.sleep(0.5)

else:
    lcd.clear()
```

```
lcd.putstr("YOU LOSE")
lcd.move_to(0, 1)
lcd.putstr("Moves: " + str(moves))
```

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main.py • diagram.json pico_i2c_lcd.py lcd_api.py

```
1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
7 I2C_ADDR = 0x27
8
9 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
10 number = 5
11 lcd.clear()
12
13 MAX_MOVES = 20
14 target_x = 10
15 target_y = 1
16
17 ball_x = random.randint(0, 15)
18 ball_y = random.randint(0, 1)
19
20 moves = 0
21 lcd.clear()
22
23 while moves < MAX_MOVES:
24     lcd.clear()
```

Simulation

00:03.147 76%

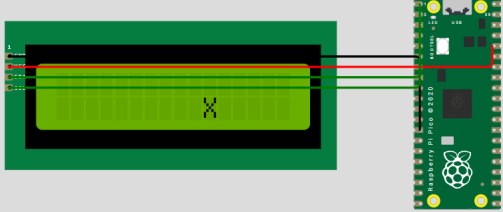
WOKWI SAVE SHARE Assignment3_6 Docs SIGN IN

main.py • diagram.json pico_i2c_lcd.py lcd_api.py

```
1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
7 I2C_ADDR = 0x27
8
9 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
10 number = 5
11 lcd.clear()
12
13 MAX_MOVES = 20
14 target_x = 10
15 target_y = 1
16
17 ball_x = random.randint(0, 15)
18 ball_y = random.randint(0, 1)
19
20 moves = 0
21 lcd.clear()
22
23 while moves < MAX_MOVES:
24     lcd.clear()
```

Simulation

00:09.393 86%



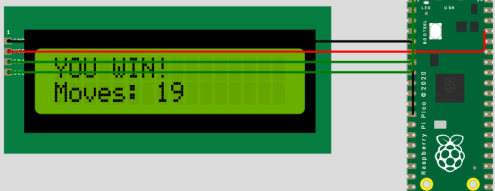
WOKWI SAVE SHARE Assignment3_6 Docs SIGN IN

main.py • diagram.json pico_i2c_lcd.py lcd_api.py

```
1 from machine import Pin, I2C
2 from pico_i2c_lcd import I2cLcd
3 import utime
4 import random
5
6 i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=40000)
7 I2C_ADDR = 0x27
8
9 lcd = I2cLcd(i2c, I2C_ADDR, 2, 16)
10 number = 5
11 lcd.clear()
12
13 MAX_MOVES = 20
14 target_x = 10
15 target_y = 1
16
17 ball_x = random.randint(0, 15)
18 ball_y = random.randint(0, 1)
19
20 moves = 0
21 lcd.clear()
22
23 while moves < MAX_MOVES:
24     lcd.clear()
```

Simulation

00:16.172 99%



MicroPython v1.24.1 on 2024-11-29; Raspberry Pi Pico with RP2040
Type "help()" for more information.
>>>